

Database Management System
Assignment- 3
Smart Blood Bank and Donor Matching System



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Screen Shots:

Smart Blood Bank And Donor Matching System

Welcome to SBBDMS

The Smart Blood Bank and Donor Matching System is a comprehensive web-based platform designed to streamline blood management by automating the connection between blood donors and recipients. The system maintains detailed digital records of donors and recipients, utilizing advanced matching logic based on blood compatibility and location to ensure timely access to life-saving resources. By integrating hospital management, appointment scheduling, and real-time inventory tracking, the project minimizes administrative overhead, reduces human error, and facilitates a rapid, reliable response mechanism during medical emergencies, ultimately enhancing the efficiency of the blood donation ecosystem.

Admin Login

Figure 1: Login Page, This App doesn't contain the sign up page as this is Admin Secured App

Smart Blood Bank And Donor Matching System

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Admin Login

Figure 2: Inserted The Username and Password, BY clicking on Login button the admin will login to the App

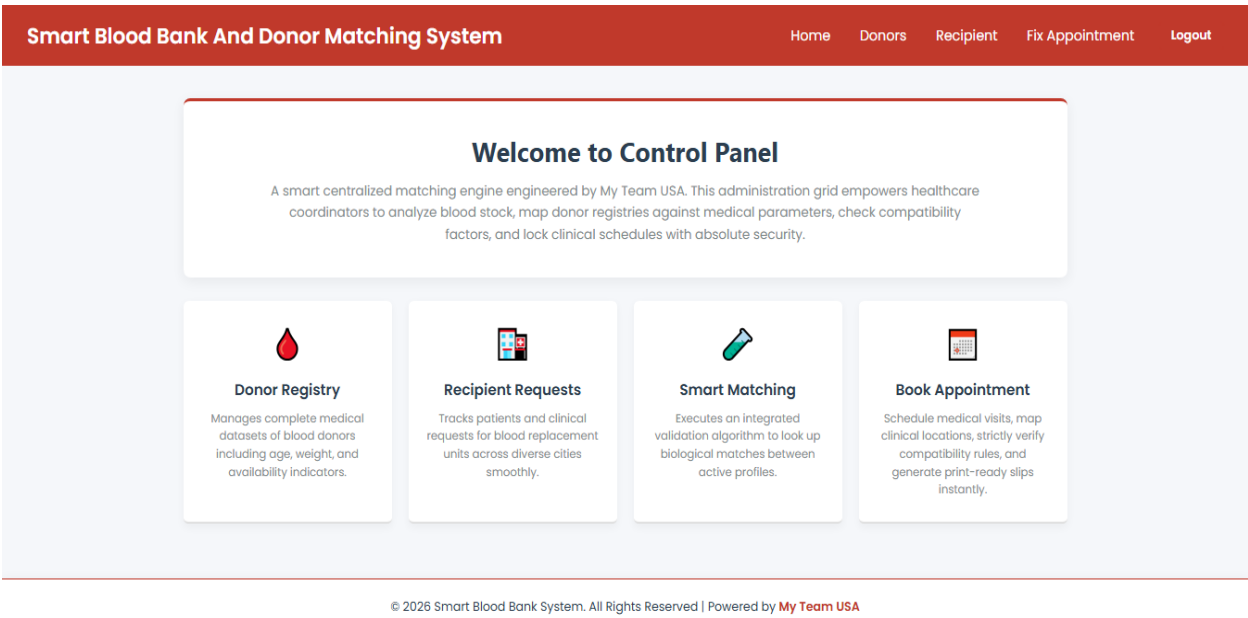


Figure 3: The Dashboard Page of the App, Showing all features and Buttons in Navigation Bar

Smart Blood Bank And Donor Matching System

HomeDonorsRecipientFix AppointmentLogout

Donor Registration & Medical Check

Full Name

CNIC (e.g. 37405-xxxxxx-x)

mm/dd/yyyy

Select Gender

Phone Number

Email Address

Blood Group (e.g. O+)

City

Age (Years)

Weight (Kg)

Hb Level (g/dL)

Select Availability

Check if Donor has any of these Medical Exclusion Conditions:

☐ Hepatitis B or C

☐ HIV/AIDS

☐ Active Infection / Fever

☐ Severe Heart Disease

☐ Uncontrolled Diabetes

Verify Constraints & Register Donor

Registered Donors (Verified List)

ID	NAME	CNIC	CONTACT	EMAIL	BO	CITY	AGE/WT	HB LEVEL	MEDICAL HISTORY	STATUS	ACTION
13	Urneeb	37403-9839864-9	0317-543212	urneeb123@gmail.com	AB+	Rawalpindi	18 yrs / 56kg	15.00 g/dL	None	Available	EditDelete
14	Umer	37202-111021112-2	0318 4240918	umerasghar426@gmail.com	AB+	Islamabad	21 yrs / 52kg	14.00 g/dL	None	Available	EditDelete

Figure 4: The Donors Page; showing the details of Donor - Personal and Medical (having Constraints), Hb level > 13, Age >18 y/o, Weight > 50 kg and must not have any of the diseases mentioned, moreover the CNIC, Phone Number and E-mail are unique.

Smart Blood Bank And Donor Matching System

[Home](#) [Donors](#) [Recipient](#) [Fix Appointment](#) [Logout](#)

Edit Donor Details (Donor ID: 13)

DONOR NAME

Urnoob

CNIC

37403-9839864-9

DATE OF BIRTH

04/02/2008

GENDER

Male

CONTACT NUMBER

0317-5432112

EMAIL ADDRESS

urnoob123@gmail.com

BLOOD GROUP

AB+

CITY

Rawalpindi

AGE (YEARS)

18

WEIGHT (kg)

56

HEMOGLOBIN (g/dl)

15.00

AVAILABILITY STATUS

Available

Update Details

Cancel

Figure 5: Edit Page for Donors

Smart Blood Bank And Donor Matching System

[Home](#) [Donors](#) [Recipient](#) [Fix Appointment](#) [Logout](#)

Recipient Management

Recipient Name

mm/dd/yyyy

Select Gender

Contact Number

Blood Group

City

Save Recipient

Recipient ID	Name	DOB	Gender	Contact	Blood Group	City	Action
2	Muaazi	2009-08-26	Male	03000000000	AB+	Rawalpindi	Edit Delete
3	Saad	2006-11-06	Male	03000000000	AB+	Lahore	Edit Delete
4	Umar	2004-07-14	Male	03000000000	O+	Lahore	Edit Delete
5	Talha	2007-07-04	Male	03000000000	A+	Rawalpindi	Edit Delete

Figure 6: Recipient Page

Smart Blood Bank And Donor Matching System

[Home](#) [Donors](#) [Recipient](#) [Fix Appointment](#) [Logout](#)

Edit Recipient Details (ID: 4)

Name

Umar

Date of Birth

07/14/2004

Gender

Male

Contact Number

03000000000

Blood Group

O+

City

Lahore

Update Recipient

Cancel

Figure 7: Edit Page for Recipient

Smart Blood Bank And Donor Matching System

[Home](#) [Donors](#) [Recipient](#) [Fix Appointment](#) [Logout](#)

Fix New Appointment

Appointment successfully scheduled! Blood groups match flawlessly.

Select Donor

Select Recipient

Select Hospital

mm/dd/yyyy

Book & Verify Compatibility

Scheduled Appointments List

APP ID	DONOR ID	DONOR BG	DONOR NAME	RECIPIENT ID	RECIPIENT BG	RECIPIENT NAME	HOSPITAL	APPOINTMENT DATE	STATUS	SLIP	ACTION
9	13	AB+	Urmoeb	3	AB+	Saad	PIMS Hospital (Pakistan Institute of Medical Sciences)	10-Jun-2026	Verified Match	Print	Edit Cancel
8	14	AB+	Umar	2	AB+	Muqazi	Kulsum International Hospital	09-Jun-2026	Verified Match	Print	Edit Cancel

Figure 8: Fix Appointment Page and also Checks the Match of Blood Groups, Can make changes, delete and print the Appointment slip

SMART BLOOD BANK AND DONOR MATCHING SYSTEM

MY TEAM USA

APPOINTMENT ID: #9

DATE: 10-Jun-2026

HOSPITAL LOCATION DETAILS

Hospital Name: PIMS Hospital (Pakistan Institute of Medical Sciences)

Hospital City: Islamabad

DONOR INFORMATION

Full Name: Urnshe

Blood Group:

Add

Contact Number: 0317-5432112

City: Rawalpindi

RECIPIENT INFORMATION

Full Name: Saad

Blood Group Required:

Add

Contact Number: 0300000000

City: Lahore

This medical match receipt is certified and issued by My Team USA. Please present this document at the designated clinical desk.

PRINT THIS SLIP

Figure 9: This our Appointment Slip and we can easily print this slip.

phpMyAdmin

Recent

Favorites

New

information_schema

mysql

performance_schema

phpmyadmin

smart_blood_bank

New

appointment

blood_stock

donation

donor

government_hospital

hospital

person

private_hospital

recipient

users

test

usa_hotel

Server: 127.0.0.1

Database: smart_blood_bank

Structure

SQL

Search

Query

Export

Import

Operations

Privileges

Routines

Events

Triggers

Tracking

Designer

Central columns

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ appointment		2	InnoDB	utf8mb4_general_ci	64.0 KiB	-
☐ blood_stock		0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ donation		0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ donor		2	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ government_hospital		0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ hospital		10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ person		13	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ private_hospital		0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ recipient		4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ users		3	InnoDB	utf8mb4_general_ci	32.0 KiB	-
10 tables	Sum	34	InnoDB	utf8mb4_general_ci	352.0 KiB	0 B

☐ Check all
 With selected:

Print

Data dictionary

Create new table

Table name

Number of columns

4

Create

Console

phpMyAdmin

Server: 127.0.0.1 > Database: smart_blood_bank > Table: appointment

Showing rows 0 - 1 (2 total. Query took 0.0373 seconds.)

SELECT * FROM 'appointment'

Number of rows: 25 Filter rows: Search this table Sort by key: None

	appointment_id	donor_id	recipient_id	hospital_id	appointment_date
☐	8	14	2	8	2026-06-06
☐	9	13	3	5	2026-06-10

Query results operations

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

phpMyAdmin

Server: 127.0.0.1 > Database: smart_blood_bank > Table: donor

Showing rows 0 - 1 (2 total. Query took 0.0007 seconds.)

SELECT * FROM 'donor'

Number of rows: 25 Filter rows: Search this table Sort by key: None

	donor_id	person_id	availability_status	hemoglobin	weight	age	major_diseases
☐	13	19	Available	15.00	56	18	None
☐	14	21	Available	14.00	52	21	None

Query results operations

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

localhost:8080/phpmyadmin/index.php?route=/table/structure&db=smart_blood_bank&table=blood_stock

phpMyAdmin

Recent Favorites

Server: 127.0.0.1 Database: smart_blood_bank Table: hospital

Showing rows 0 - 9 (10 total. Query took 0.0006 seconds.)

SELECT * FROM `hospital`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	hospital_id	hospital_name	city	contact_number
☐ Edit ☐ Copy ☐ Delete	4	Shifa International Hospital	Islamabad	051-8453000
☐ Edit ☐ Copy ☐ Delete	5	PIMS Hospital (Pakistan Institute of Medical Scien...	Islamabad	051-9261170
☐ Edit ☐ Copy ☐ Delete	6	Holy Family Hospital	Rawalpindi	051-9290321
☐ Edit ☐ Copy ☐ Delete	7	Benazir Bhutto Hospital (BBH)	Rawalpindi	051-9290301
☐ Edit ☐ Copy ☐ Delete	8	Kulsum International Hospital	Islamabad	051-8445696
☐ Edit ☐ Copy ☐ Delete	9	Quaid-e-Azam International Hospital	Islamabad	051-8449100
☐ Edit ☐ Copy ☐ Delete	10	Armed Forces Bone Marrow Transplant Centre (AFBMTC...	Rawalpindi	051-9273295
☐ Edit ☐ Copy ☐ Delete	11	Fauji Foundation Hospital	Rawalpindi	051-5154311
☐ Edit ☐ Copy ☐ Delete	12	Maroof International Hospital	Islamabad	051-111644911
☐ Edit ☐ Copy ☐ Delete	13	Rawalpindi General Hospital (RGH / Bilal Hospital)	Rawalpindi	051-4456111

Check all With selected: Edit Copy Delete Export

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

localhost:8080/phpmyadmin/index.php?route=/server/databases

phpMyAdmin

Recent Favorites

Server: 127.0.0.1 Database: smart_blood_bank Table: person

Showing rows 0 - 12 (13 total. Query took 0.0021 seconds.)

SELECT * FROM `person`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

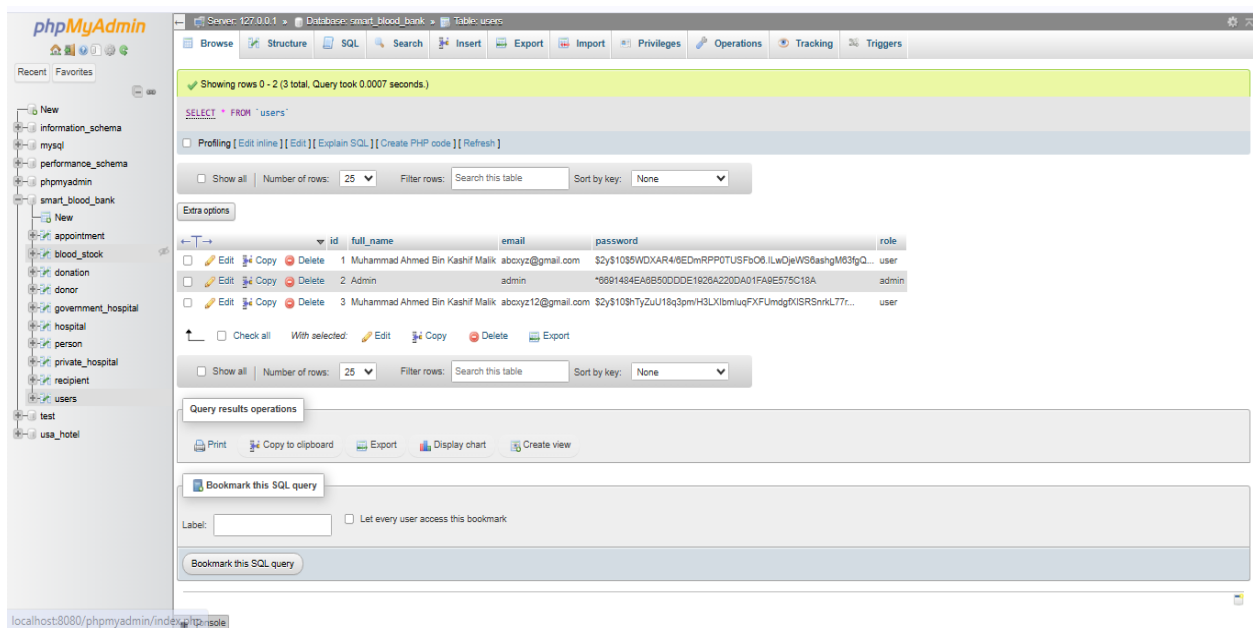
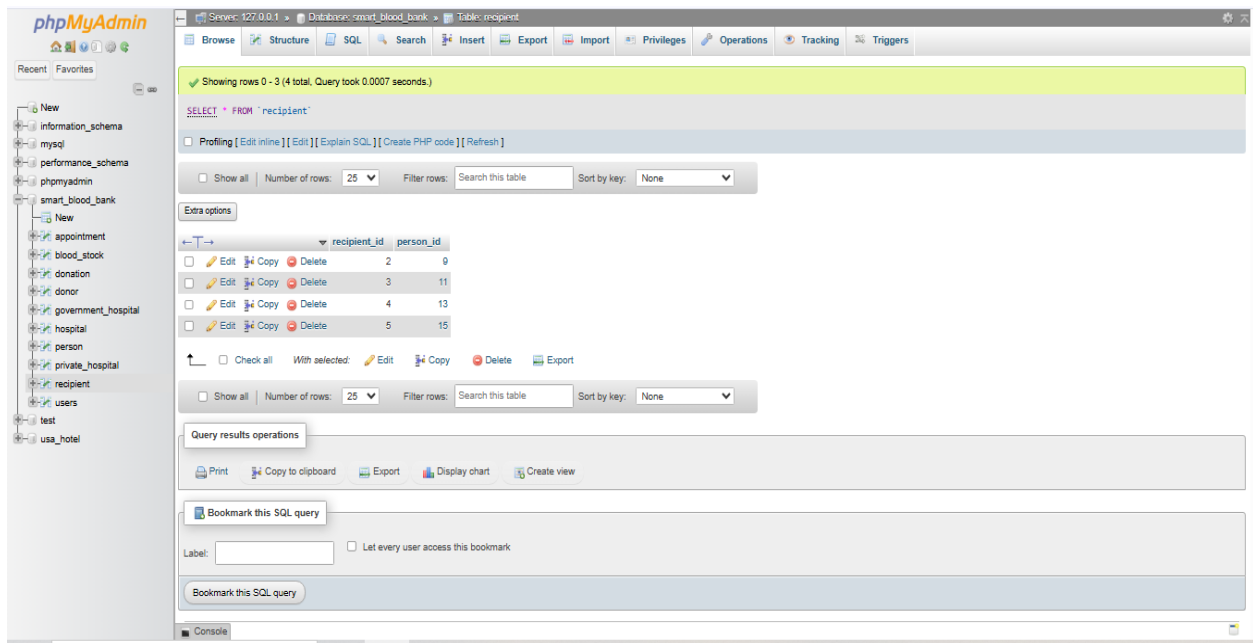
Extra options

	person_id	name	cnic	date_of_birth	gender	contact_number	email	city	blood_group
☐ Edit ☐ Copy ☐ Delete	6	Ahmed		2006-11-22	Male	03000000000		Rawalpindi	AB+
☐ Edit ☐ Copy ☐ Delete	7	Saad		2006-09-18	Male	03000000000		Rawalpindi	B-
☐ Edit ☐ Copy ☐ Delete	9	Muaazi		2009-08-26	Male	03000000000		Rawalpindi	AB+
☐ Edit ☐ Copy ☐ Delete	10	Umar		2000-08-20	Male	03000000000		Rawalpindi	O+
☐ Edit ☐ Copy ☐ Delete	11	Saad		2006-11-06	Male	03000000000		Lahore	AB+
☐ Edit ☐ Copy ☐ Delete	12	Sanim		2007-03-29	Male	03000000000		Islamabad	A+
☐ Edit ☐ Copy ☐ Delete	13	Umar		2004-07-14	Male	03000000000		Lahore	O+
☐ Edit ☐ Copy ☐ Delete	15	Talha		2007-07-04	Male	03000000000		Rawalpindi	A+
☐ Edit ☐ Copy ☐ Delete	16	Amir		2007-04-24	Male	03000000000		Karachi	B-
☐ Edit ☐ Copy ☐ Delete	17	ahmad Asghar		2026-06-08	Male	03226967548		Islamabad	AB+
☐ Edit ☐ Copy ☐ Delete	18	ahmad Asghar	37202-6689762-2	2002-07-08	Male	03226967554	ahmedasghar426@gmail.com	Islamabad	AB+
☐ Edit ☐ Copy ☐ Delete	19	Urmeeb	37403-9838084-9	2008-04-02	Male	0317-5432112	urmeeb123@gmail.com	Rawalpindi	AB+
☐ Edit ☐ Copy ☐ Delete	21	Umer	37202-111021112-2	2003-11-28	Male	0316 4240918	umerasghar426@gmail.com	Islamabad	AB+

Check all With selected: Edit Copy Delete Export

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Console/SQL operations



Working Explanation

Step 1: Installing XAMPP

XAMPP server was installed to run Apache and MySQL services locally on the system.

Step 2: Creating Database

The project database was created using phpMyAdmin. Required normalized tables were created using SQL queries.

Step 3: Connecting Database with Frontend

PHP was used to connect the frontend with the MySQL database using mysqli connection.

Step 4: Implementing CRUD Operations

CRUD operations were implemented:

- Create → Add new records
- Read → Display records
- Update → Edit existing records
- Delete → Remove records

Step 5: Testing

All pages were tested successfully and database records were properly stored and retrieved from MySQL server.